PARISHRAM 2.0

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS | LECTURE-10

BY-DEV AGNANI SIR



Recap

Syllabus

1. Electric Field due to Uniformly Charged thin Spherical Shell

2. Electric Field due to a Non-Conducting Charged Solid Sphere

Not in syllabus

Syllabus

3. Electric Field due to a Uniformly Charged Infinite Plane Sheet

4.


Muh chhupake **nikalte** huye
Dev sir after his **prank call**
on **Mandar sir** got failed:

Topics to be covered:

1. Electric Dipole and Dipole Moment

2. Electric Field due to an Electric Dipole at:
   1. Axial Line
   2. Equatorial Line

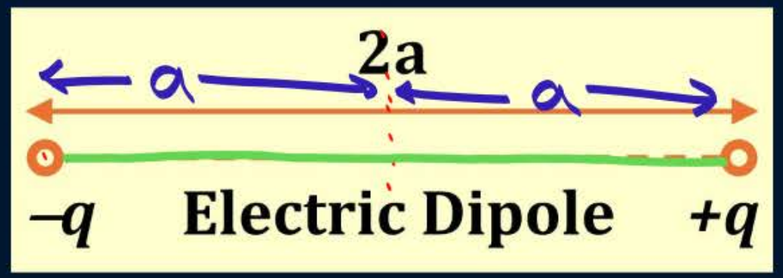
3. Electric Dipole in a Uniform Electric Field

Basic maths
- → Differentiation
- → Integration
- → Vectors
- → Trigonometry
- → Straight Line
- → Areas and Volumes.

Electric Dipole

An **electric dipole** is described as a **pair of equal** and **opposite** point charges, denoted as **q** and **–q**, separated by a **small distance** which is referred to as **2a.**

length of Dipole = 2a



Dipole moment is a vector quantity.

↳ Direction: –q ⟶ +q from -ve charge to +ve charge.

Dipole Moment

$$\vec{P} = q(2\vec{a})$$

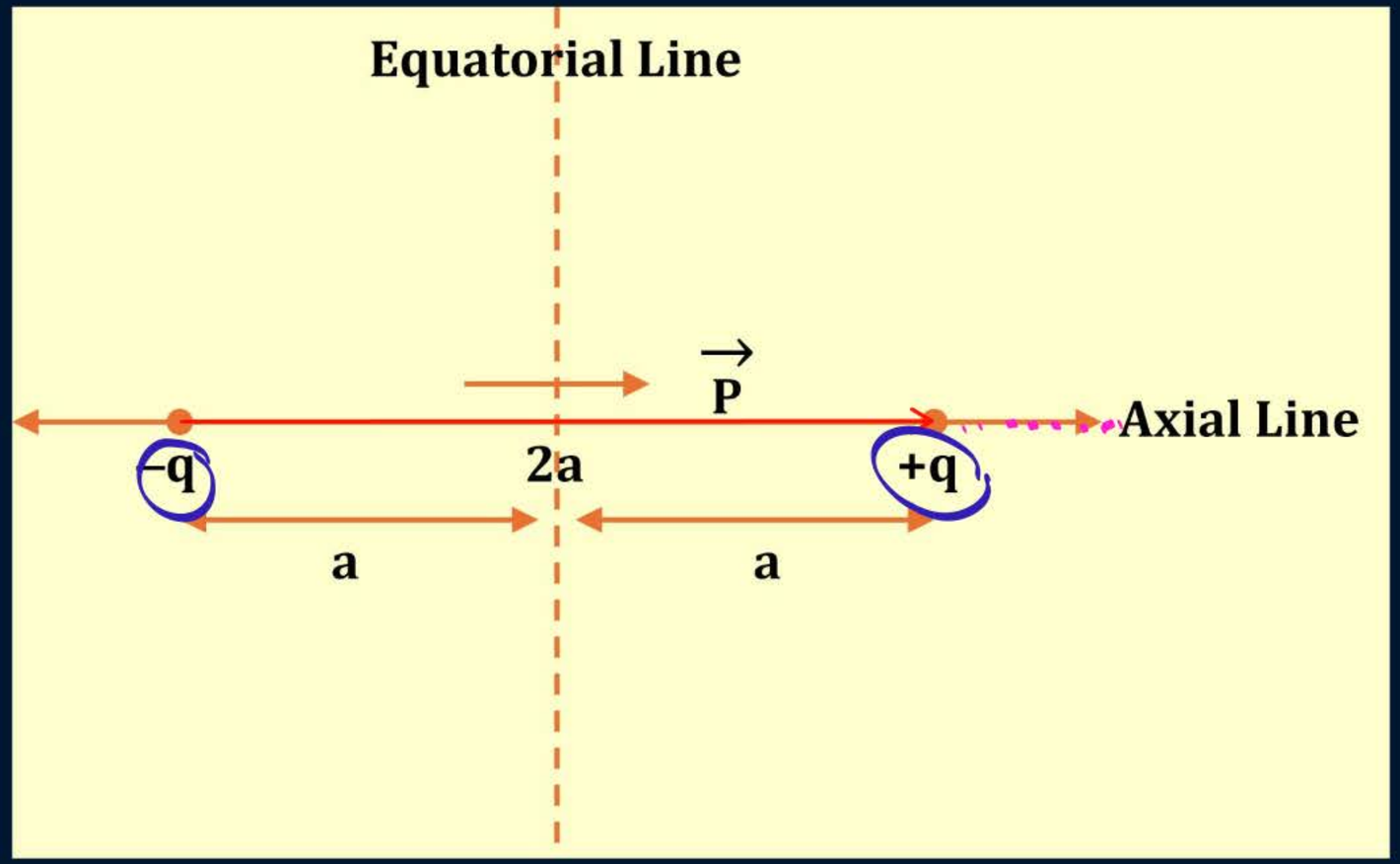
It is defined as the product of **magnitude** of either charge and the **distance** between the **two** charges.

$$\vec{P} = q\left(2\vec{a}\right)$$

Dipole moment always $\vec{P}$ points from **–q** to **+q**.

S.I. unit of Dipole moment ($\vec{p}$)

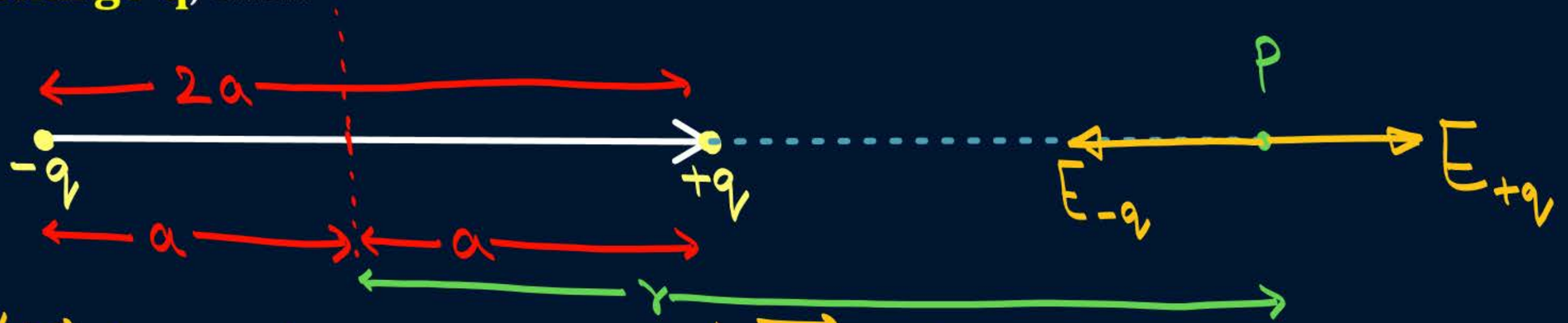
⇒ Cm



Electric Field due to an Electric Dipole for points on the axis

(for point on the axial line)

Let the **point P** be at **distance r** from the centre of the **dipole** on the side of the **charge q**, then



$$(\vec{E}_{net})_P = (\vec{E}_{+q}) + (\vec{E}_{-q})$$

$$\vec{E}_{+q} = \frac{kq}{(r-a)^2}(\hat{p})$$

$$\vec{E}_{-q} = \frac{kq}{(r+a)^2}(-\hat{p})$$

unit vector of Dipole :-

$$\hat{p} = \frac{\vec{p}}{|\vec{p}|}$$

to tell the direction.

$$\vec{E}_{net} = \frac{Kq}{(r-a)^2}(\hat{p}) + \frac{Kq}{(r+a)^2}(-\hat{p})$$

$(a+b)^2 = a^2+b^2+2ab$
$(a-b)^2 = a^2+b^2-2ab$
$(a+b)(a-b) = a^2-b^2$

$$= Kq\left\{\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right\}\hat{p} = Kq\left\{\frac{r^2+a^2+2ra-r^2-a^2+2ra}{(r-a)^2(r+a)^2}\right\}\hat{p}$$

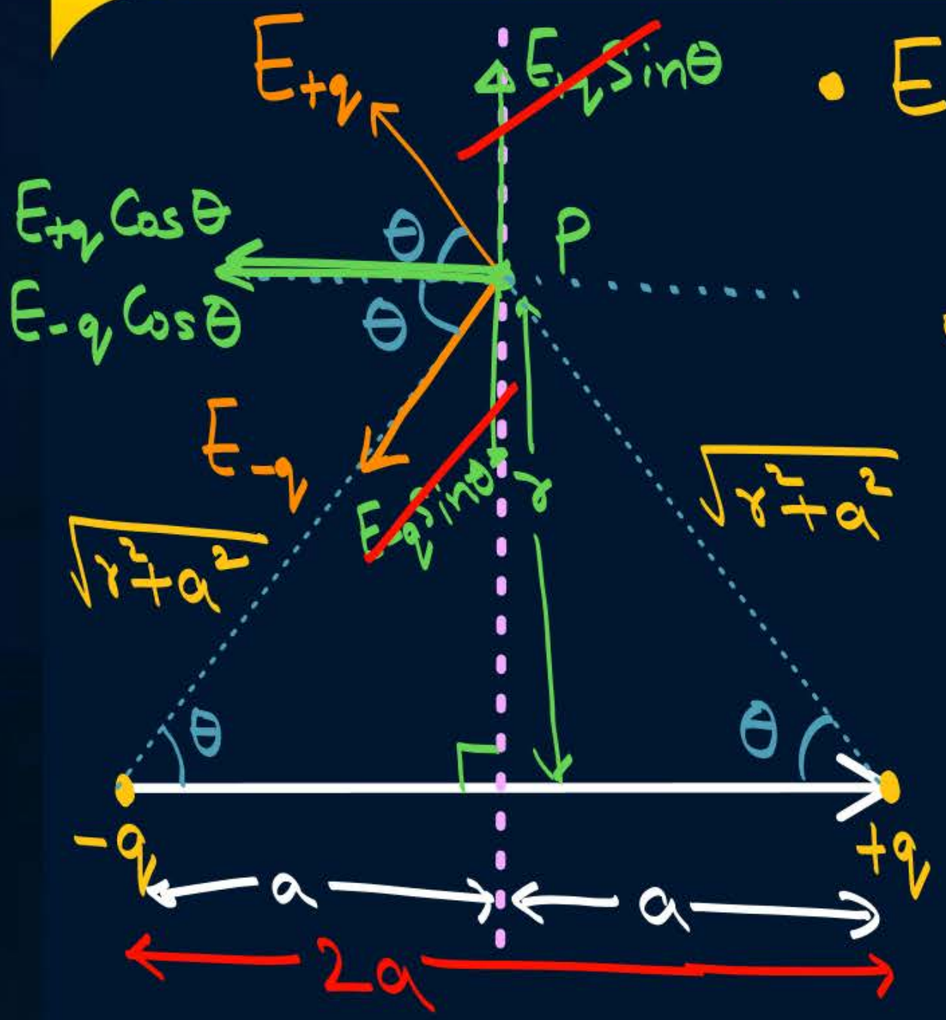
$$= Kq\left[\frac{4ra}{(r^2-a^2)^2}\right]\hat{p}$$

if $r >> a$, then $r^2 >>>> a^2$;

$$= \frac{2Kq(q2a)}{r^3}\hat{p} = \frac{2K(\vec{p})}{r^3}\left(\frac{\vec{r}}{|\vec{r}|}\right) \Rightarrow \boxed{\vec{E} = \frac{2K\vec{p}}{r^3}}$$

E.F. due to a Dipole on a point on the axial line.

Electric Field due to an Electric Dipole for points on the Equatorial plane {Equitorial line always bisects the dipole}.



- $E_{+q} = \dfrac{kq}{(\sqrt{r^2+a^2})^2} \Rightarrow |E_{+q}| = \dfrac{kq}{r^2+a^2}$
- $|E_{-q}| = \dfrac{kq}{r^2+a^2}$

$$\vec{E}_{net} = E_{+q}\cos\theta(-\hat{p}) + E_{-q}\cos\theta(-\hat{p})$$

$$= (E_{+q} + E_{-q})\cos\theta(-\hat{p})$$

$$= \frac{2kq}{(r^2+a^2)} \times \frac{a}{\sqrt{r^2+a^2}}(-\hat{p})$$

$$\vec{E}_{net} = \frac{K(q2a)}{(r^2+a^2)} \times \frac{1}{(r^2+a^2)^{1/2}} (-\hat{p}) = \frac{K|\vec{p}|}{(r^2+a^2)^{3/2}} \left(\frac{-\vec{p}}{|\vec{p}|}\right)$$

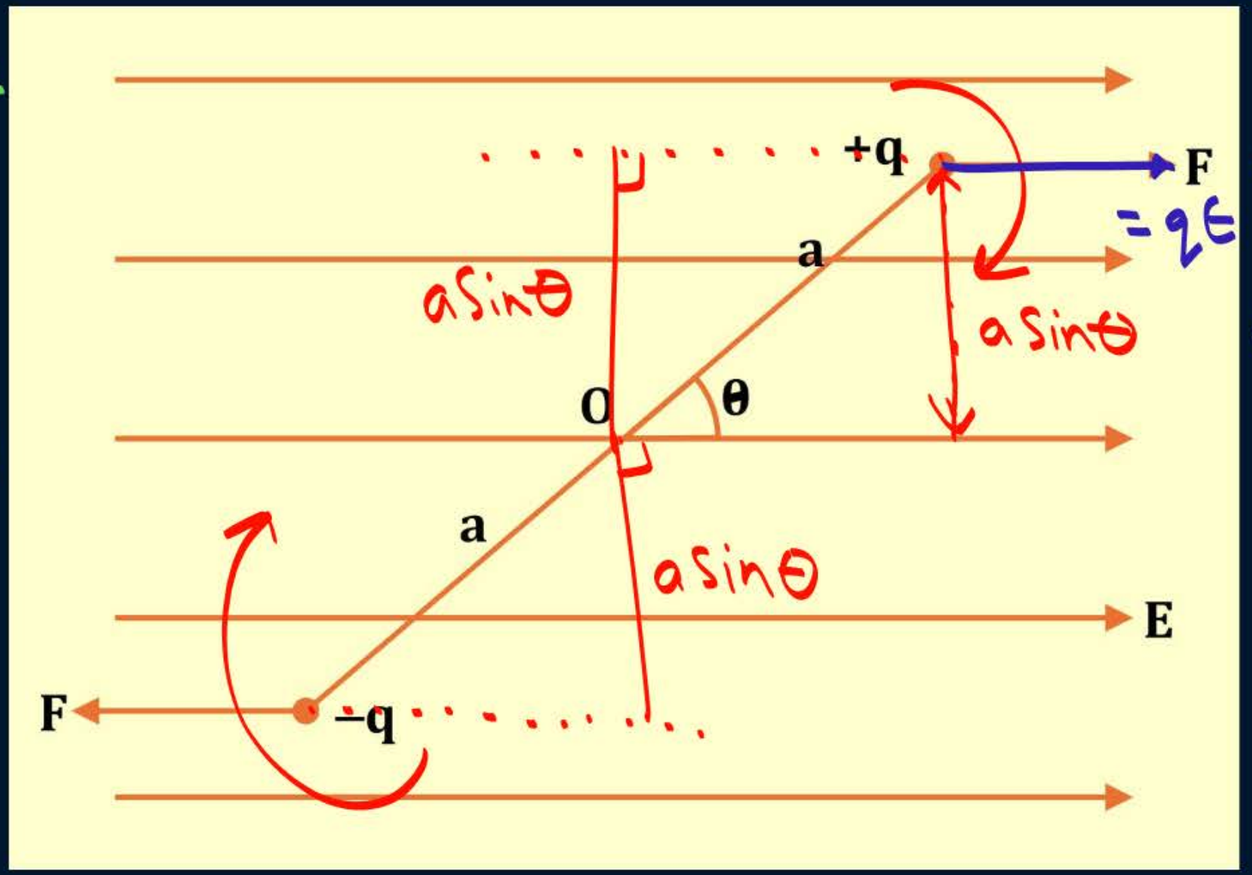
$$\Rightarrow \vec{E} = \frac{-K\vec{p}}{(r^2+a^2)^{3/2}} \quad ; \quad r >> a \quad ; \quad r^2 >>>> a^2 \quad ;$$

$$\Rightarrow \boxed{\vec{E} = \frac{-K\vec{p}}{r^3}}$$

→ E.F. due to a Dipole on a point on the equitorial line.

Electric dipole in a uniform electric field

When a Dipole $\vec{p}$ made up of two equally and opposite charges $+q$ and $-q$ seperated by distance $2a$, is placed in a Uniform E.F. at an angle θ to the Electric field lines, then the Dipole will experience a torque on it.



$\star\ \tau_{+q} = F\,a\sin\theta$

$\tau_{+q} = qE\,a\sin\theta$

$\Rightarrow \tau_{+q} = qaE\sin\theta$

$\star\ \tau_{-q} = F\,a\sin\theta$

$\Rightarrow \tau_{-q} = qaE\sin\theta$

$$\tau_{net} = \tau_{+q} + \tau_{-q}$$

$$\tau_{net} = \underbrace{2aq}_{p} E \sin\theta$$

$$\Rightarrow \tau_{net} = pE\sin\theta$$

$$\Rightarrow \boxed{\vec{\tau}_{net} = \vec{p} \times \vec{E}}$$

Dot Product :- → Scalar Product

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$

Cross Product :- → Vector Product

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta \, \hat{n}$$

Special Cases :-

(I) $\theta = 0^\circ$
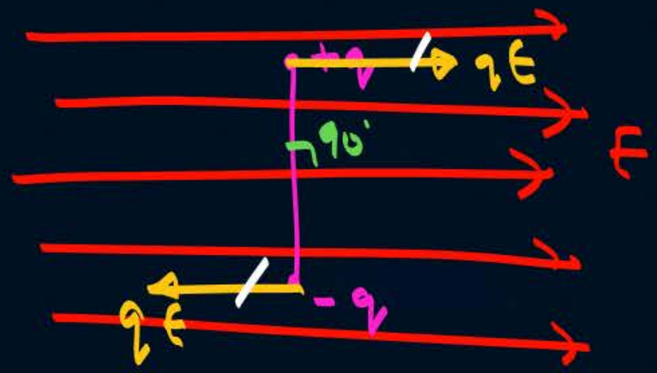


$\tau = pE\sin\theta$

$\Rightarrow \boxed{\tau = 0}$.

$\Rightarrow \boxed{F_{net} = 0}$.

(II) $\theta = 90^\circ$
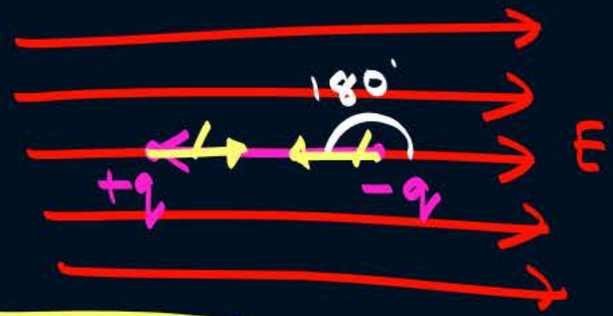


$\Rightarrow \boxed{f_{net} = 0}$

$\tau = pE\sin\theta$

$\Rightarrow \boxed{\tau = pE}$

(III) $\theta = 180^\circ$



$\boxed{F_{net} = 0}$

$\tau = pE\sin(180^\circ)$

$\Rightarrow \boxed{\tau = 0}$.

QUESTION

Torque acting on a dipole in a **uniform electric field** is maximum when the angle between dipole **moment** and **field** is:

$\tau = pE\sin\theta$

A 0°

B 45°

C 90°

D 180°



QUESTION

The **magnitude** of **electric field** on the axial line compared to **equatorial line** (for the same distance r) is:

Axial line	Equitorial line
$\vec{E} = \frac{2k\vec{p}}{r^3}$	$\vec{E} = -\frac{k\vec{p}}{r^3}$

A Double

B Half

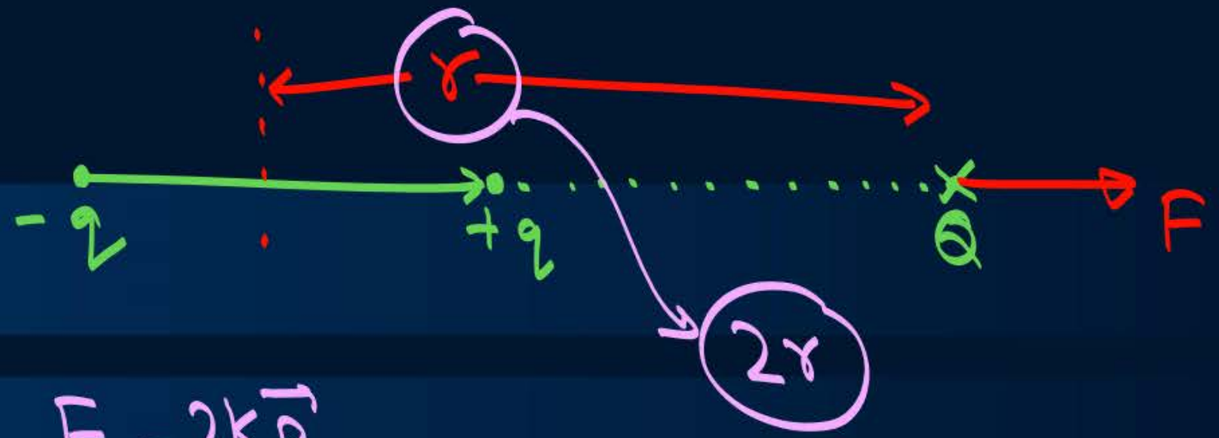
C Same

D Four times.



QUESTION

The **force** on a charge **situated** on the **axis** of a **dipole** is **F**, if the charge is **shifted** to **double** the distance, the acting **force** will be



$E = \frac{2kp}{r^3}$



A Zero

B $\frac{F}{2}$

C $\frac{F}{4}$

D $\frac{F}{8}$

QUESTION

HW

An **electric dipole**, when held at **30°** with respect to a uniform **electric field** of **10^4 N/C** experiences a torque of **9×10^{-26} Nm**. Calculate dipole **moment** of the **dipole**

A p = 0.9 x 10^-28 Cm

B p = 1.8 x 10^-28 Cm

C p = 1.8 x 10^-29 Cm

D p = 0.9 x 10^-29 Cm



QUESTION

V.V. imp. ques.

The **magnitude** of **dipole moment** of the following system is:

$$P_R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2\cos\theta}$$

$$= \sqrt{(qa)^2 + q^2a^2 + 2(qa)(3qa)\left(\frac{1}{2}\right)}$$

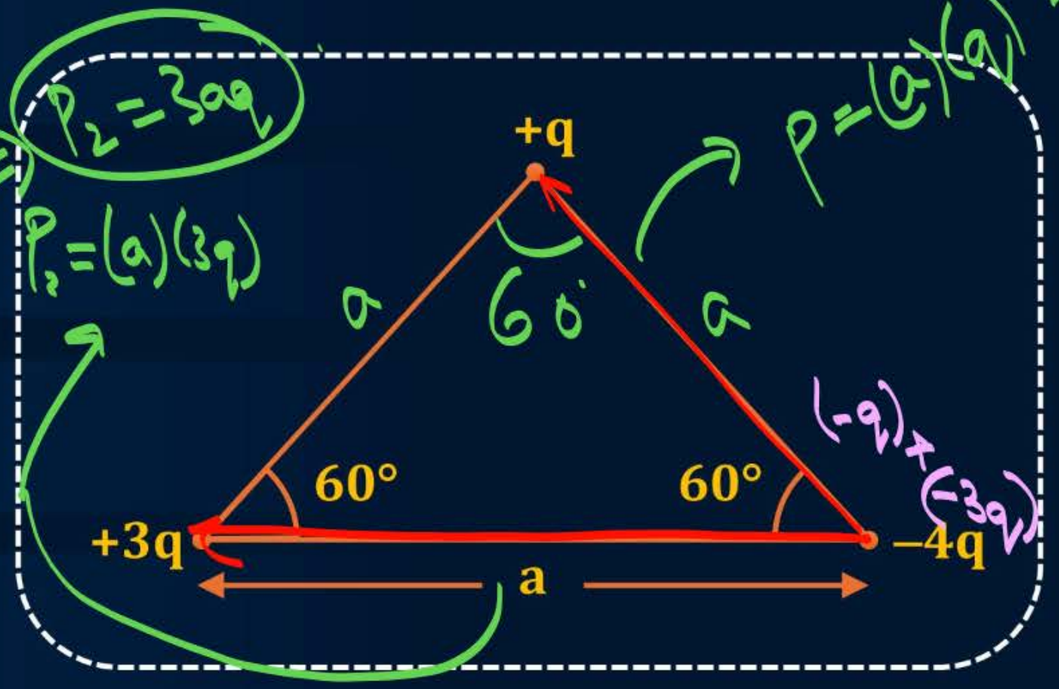
$$= \sqrt{13}\,qa$$

A. $qa\sqrt{21}$

B. $qa\sqrt{13}$

C. 4qa

D. Zero



QUESTION

Find the magnitude of **dipole moment** of the following system.

$$P_R = \sqrt{4q^2a^2 + 9q^2a^2 + 2(2qa)(3qa)\left(\frac{1}{2}\right)}$$

$$P_R = \sqrt{19q^2a^2}$$

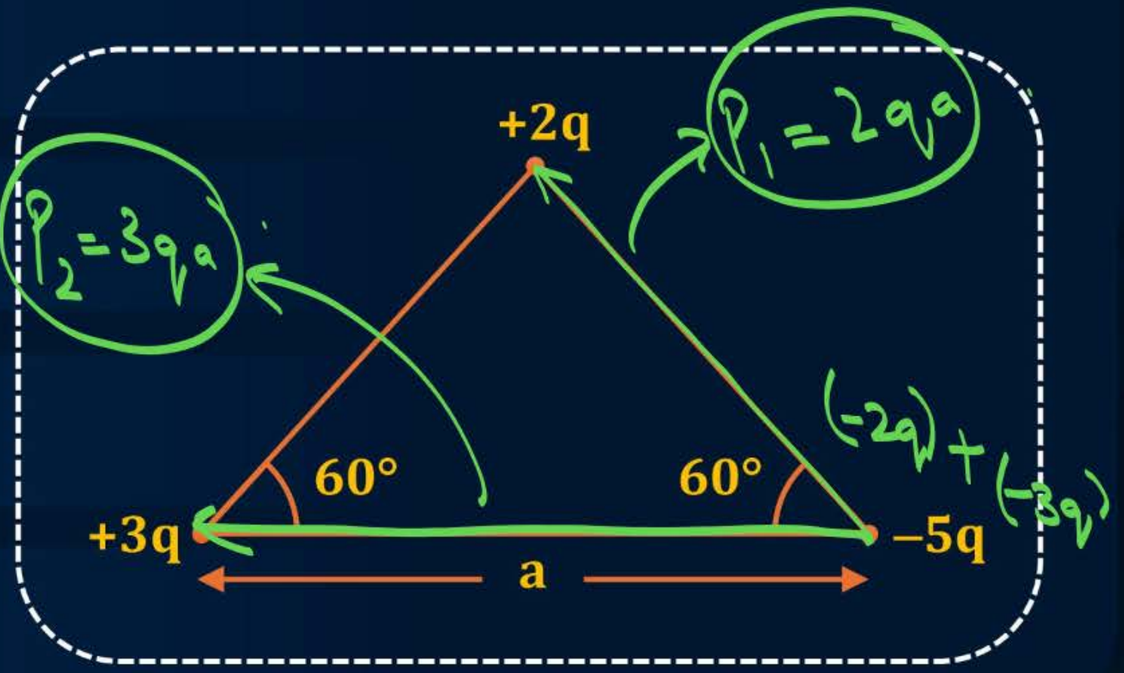
$$P_R = \sqrt{19}\,qa$$

A $qa\sqrt{21}$

B $qa\sqrt{13}$

C $qa\sqrt{19}$

D Zero



QUESTION

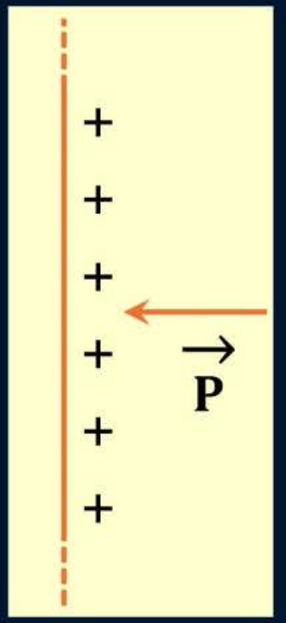
HW. An **electric dipole** is placed **perpendicular** to an infinite line of charge at small distance as shown in **figure**. Identify the correct statement.

A The dipole is attracted towards the line charge

B The dipole is repelled away from the line charge

C The dipole does not experience a force

D The dipole experience a force as well as a torque.



QUESTION

(4) An **electric dipole** of dipole moment consists of point charges **q** and **–q**, separated by **2a**. Derive an expression for **electric potential** in terms of its dipole moment at a point at a distance **x(>> a)** from its centre and lying

(I) along its **axis**, and

(II) along its **bisector line**.

(H.W.)

QUESTION

(H)

An **electric** dipole of dipole moment $\vec{p} = (0.8\hat{i} + 0.6\hat{j})10^{-29}$ **cm** is placed in an electric field $\vec{E} = 1.0 \times 10'\hat{k}\frac{V}{m}$. Calculate magnitude of the **torque acting** on it and the angle it makes with the **x-axis**, at this instant.

10 lectures → CHAPTER समाप्त !

Ques. Practise.

- NCERT
- S.L. Arora
- DPP
- Internet.

THANK YOU

